TEXORA · COURSE NOTES · LECTURE SHEET

Theories of Co-ordination Chemistry

Complex compounds from Werner's insight to crystal field theory — ligands, chelates, naming, isomerism, magnetism and colour.

CHE-1105 · CHEMISTRY-I · TOPIC 4 · LEVEL 1 TERM 1

1. What a complex is

Mix cobalt chloride with ammonia and the product, $CoCl_3 \cdot 6NH_3$, refuses to behave like a mere mixture: only three of its chlorides precipitate with silver nitrate, and its conductivity is that of a salt with four ions. The explanation is the **coordination compound**: a central metal ion (a Lewis acid) surrounded by molecules or ions called **ligands** (Lewis bases) that donate electron pairs into its empty orbitals. The metal and ligands together form the **coordination sphere**, written in square brackets — $[Co(NH_3)_6]Cl_3$ — and anything outside the brackets is ionisable, ordinary salt. The number of donor atoms attached to the metal is its **coordination number** (6 here, the commonest, giving octahedral geometry).

2. Werner's theory

Alfred Werner (1893) explained the cobalt ammines with two kinds of valency:

- **Primary valency** — the metal's oxidation state, satisfied by anions, ionisable (the three Cl^- that precipitate as AgCl).
- **Secondary valency** — the coordination number, satisfied by ligands, non-ionisable and *directed in space*: six secondary valencies point to the corners of an octahedron, four to a square plane or tetrahedron.

The series $[Co(NH_3)_6]Cl_3$, $[Co(NH_3)_5Cl]Cl_2$, $[Co(NH_3)_4Cl_2]Cl$ falls out immediately: as NH_3 ligands are replaced by Cl^- *inside* the sphere, one fewer chloride remains ionisable outside. Werner predicted, before they were found, the two forms of $[Co(NH_3)_4Cl_2]^+$ — a structural triumph that won him the Nobel Prize and founded the subject.

3. Ligands and chelation

Ligands are classified by how many donor atoms they attach through: **monodentate** (one: NH_3, H_2O, Cl^-, CN^-), **bidentate** (two: ethylenediamine "en", oxalate), and **polydentate** (several: EDTA attaches

through six donor atoms). A polydentate ligand wrapping a metal in a ring is a **chelate** (Greek: a claw), and chelates are markedly more stable than comparable monodentate complexes — the **chelate effect**, driven largely by entropy: one EDTA displaces six water molecules, increasing disorder. Chelation is practical chemistry: EDTA sequesters Ca^{2+} and metal ions in water softening and in dyeing (where stray metal ions spoil shades), and chelated metals serve as drugs, catalysts and pigments.

4. Naming complexes (IUPAC essentials)

- Ligands are named first, alphabetically, with multiplying prefixes (di-, tri-, tetra-; bis-, tris- for complicated names), then the metal, then the oxidation state in Roman numerals.
- Anionic ligands end in -o: chloro, cyano, hydroxo, oxalato. Neutral ligands keep their names, with the famous exceptions aqua (H_2O) and ammine (NH_3).
- If the complex itself is an anion, the metal takes the ending -ate (ferrate, cobaltate, platinate).

Examples: $[Co(NH_3)_6]Cl_3$ is hexaamminecobalt(III) chloride; $K_4[Fe(CN)_6]$ is potassium hexacyanoferrate(II); $[Pt(NH_3)_2Cl_2]$ (the anti-cancer drug cisplatin in its cis form) is diamminedichloroplatinum(II).

5. Isomerism

Complexes of identical composition can differ structurally. **Ionisation isomerism**: $[Co(NH_3)_5SO_4]Br$ and $[Co(NH_3)_5Br]SO_4$ exchange a ligand with an outside ion. **Linkage isomerism**: a ligand binds through different atoms (NO_2 through N or O). **Geometrical isomerism**: in square-planar MA_2B_2 and octahedral MA_4B_2 complexes, like ligands may be adjacent (*cis*) or opposite (*trans*) — Werner's predicted pair; cisplatin's anti-cancer activity vanishes in its trans form, a standing lesson that geometry is function. **Optical isomerism**: tris-chelate octahedral complexes such as $[Co(en)_3]^{3+}$ are chiral propellers, existing as non-superimposable mirror images.

6. Valence bond theory of complexes

Pauling's valence bond approach lets magnetism arbitrate structure. In $[Fe(CN)_6]^{4-}$ the ion is **diamagnetic** — all electrons paired. Fe^{2+} is d^6; to pair six electrons into three 3d orbitals, two 3d orbitals must empty for bonding: the complex uses two 3d, one 4s and three 4p orbitals — **d^2sp^3, an inner-orbital complex** — and is octahedral with no unpaired electrons. In $[FeF_6]^{3-}$ the ion is strongly **paramagnetic** with five unpaired electrons: Fe^{3+} (d^5) keeps every 3d electron unpaired, bonding must use outer 4d orbitals — **sp^3d^2, an outer-orbital complex**. Ligands that force pairing (CN^-, CO) are now called strong-field; those that do not (F^-, H_2O) weak-field — the observation crystal field theory explains.

7. Crystal field theory

Crystal field theory treats ligands as point charges approaching a metal ion. In an octahedral complex the six ligands approach along the axes, so the d orbitals split into two sets: the two orbitals pointing *along* the axes (d_{z^2}, $d_{x^2-y^2}$, the e_g set) are repelled upward in energy, and the three pointing *between* axes (d_{xy}, d_{yz}, d_{zx}, the t_{2g} set) drop lower; the gap is $\boldsymbol{\Delta_o}$. Filling then balances Δ_o against the pairing energy: strong-field ligands (large Δ_o: CO, CN^-, en, NH_3) force electrons to pair in t_{2g} — low-spin complexes; weak-field ligands (small Δ_o: I^-, Br^-, Cl^-, F^-, H_2O) let them spread — high-spin complexes. The ligand order is the **spectrochemical series**: $I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 <$ en $< CN^- <$ CO. Two consequences are visible to the naked eye: **colour** — complexes absorb light by promoting electrons across Δ_o, so the colour seen is the complement of the colour absorbed, and changing a ligand changes Δ_o and the colour; and **magnetism**, read directly off the electron filling. Crystal field theory thus explains in one diagram what Werner catalogued and Pauling half-explained — and with ligand field theory (its covalent refinement) it remains the working theory of transition-metal colour, from dyestuffs to the blue of copper sulphate.

8. Glossary — key terms of this topic

TERM	MEANING
Ligand	A molecule or ion donating an electron pair to the central metal.
Coordination number	The number of donor atoms bonded to the central metal.
Chelate	A complex in which a polydentate ligand forms a ring (or rings) with the metal.
Primary / secondary valency	Werner’s terms for oxidation state (ionisable) and coordination number (non-ionisable, directional).
High-spin / low-spin	Complexes with maximum unpaired electrons (weak field) or forced pairing (strong field).
Δ_o	The crystal field splitting energy between the t_{2g} and e_g orbital sets in an octahedral field.
Spectrochemical series	The ordering of ligands by the size of splitting they produce.

9. Key points and summary

- A complex is a metal ion with ligands in a coordination sphere; Werner’s primary/secondary valencies explain ionisable versus bound ions and predict geometry and isomers.

- Ligands are counted by denticity; chelates gain extra stability (the chelate effect) — the basis of EDTA's use in water treatment and dyeing.
- Valence bond theory reads structure from magnetism (inner d^2sp^3 vs outer sp^3d^2); crystal field theory explains the same facts, plus colour, through the t_{2g}/e_g splitting Δ_o and the spectrochemical series.

10. Practice

Answer in your own words; brief model answers follow.

1. For $[Co(NH_3)_5Cl]Cl_2$: state the oxidation state of cobalt, its coordination number, and how many moles of AgCl precipitate per mole of compound with excess $AgNO_3$.
2. Name $K_4[Fe(CN)_6]$ and $[Co(NH_3)_5Cl]Cl_2$ by the rules of §4.
3. $EDTA^{4-}$ displaces six water molecules from $[Fe(H_2O)_6]^{3+}$. Name the effect responsible for the exceptional stability of the product and its driving force.
4. Draw (in words) the cis and trans forms of square-planar $[Pt(NH_3)_2Cl_2]$, and state which is the drug cisplatin.
5. $[Fe(CN)_6]^{4-}$ is diamagnetic; $[FeF_6]^{3-}$ has five unpaired electrons. Using §6, give each complex's hybridisation and inner/outer-orbital label.
6. Place these ligands in order of increasing Δ_o: NH_3, F^-, CN^-, H_2O — then predict which gives the lowest-spin complex with a d^6 metal.
7. A complex absorbs orange light strongly. What colour does it appear, and what transition absorbs the light?
8. Werner's secondary valency of $[Co(NH_3)_4Cl_2]^+$ is 6. What geometry follows, and how many geometrical isomers does the complex have?

Answer key

1. Oxidation state +3; coordination number 6 (5 NH_3 + 1 Cl inside the sphere); only the 2 outside chlorides precipitate — 2 moles of AgCl.
2. Potassium hexacyanoferrate(II); pentaamminechlorocobalt(III) chloride.
3. The chelate effect — driven mainly by entropy: one EDTA replaces six waters, so the number of free particles increases sharply.
4. Cis: the two Cl adjacent (90° apart); trans: opposite (180°). The cis form is cisplatin.
5. $[Fe(CN)_6]^{4-}$: d^2sp^3, inner-orbital (all d^6 electrons paired). $[FeF_6]^{3-}$: sp^3d^2, outer-orbital (d^5 all unpaired).
6. $F^- < H_2O < NH_3 < CN^-$. CN^- gives the largest Δ_o and hence the low-spin complex.

7. Blue (the complement of orange). The absorbed light promotes an electron from t_{2g} to e_g — a d–d transition of energy Δ_o.
8. Octahedral. Two — cis and trans (the pair Werner predicted).
